



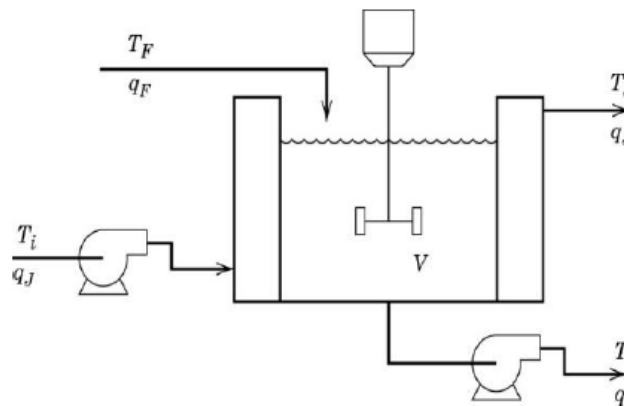
UNIVERSITY OF CAPE TOWN
DEPARTMENT OF CHEMICAL ENGINEERING
CHE3007S BLOCK 3 DATA SCIENCE

PROJECT BRIEF: DATA ANALYSIS OF A NON-ISOTHERMAL CSTR SYSTEM

INTRODUCTION

An irreversible exothermic reaction, where A forms species B , is carried out in a CSTR.

In the schematic shown below, the inlet stream to the reactor comprises pure component A with molar concentration, c_{Ai} . Since the reaction is exothermic, the reactor is surrounded by a cooling jacket whose purpose is to maintain the reaction mixture at the desired operating temperature by absorbing heat released by reaction.



Data has been collected over the span of one month. Most of the data is automatically generated through high-frequency sensors, apart from concentration measurements, which are conducted in a laboratory setting. The reactor aims to maintain an effluent concentration of A below 0.1 mol/L. Any deviation from this target incurs a penalty of \$2 USD per 0.01 mol/L over the target concentration.

Your line manager has entrusted you with the task of utilizing this data to gain deeper insights into the reactor's operation and provide recommendations for enhancing its performance. Your output will be a concise report (maximum 3 pages) for your line manager, who may have limited exposure to data-driven analysis.

TOTAL MARKS:65

- 1) Process Understanding [5]
 - a. Generate a system diagram of the process followed by a discussion of which variables are disturbances, manipulated variables, outputs and states. (5)
- 2) Data pre-processing [11]
 - a. Identify and discuss any concerns or issues with the data. (3)
 - b. Perform data cleaning and justify all the techniques used by discussing them. (8)
- 3) Model development and evaluation [34]
 - a. Transform and partition your data appropriately, providing justifications for your decisions in this process. (4)
 - b. Train linear regression, K-nearest neighbours, and neural network models to predict your KPI (concentration of A in your effluent). *Do not do hyperparameter selection at this point. Note that you will use this model for prescriptive analytics later, so you should not include any dependent variables other than your KPI.* (10)
 - c. Choose and justify the model to advance based on its performance in predicting the validation data, using appropriate performance metrics. Also, comment on the hyperparameters of the selected model. (10)
 - d. Evaluate the model's performance on the test data, identify key input variables, and comment on both the model's performance and whether the identified variables align with your domain knowledge. (10)
- 4) Recommendations [15]
 - a. Due to upgrades on the cooling water circuit, the temperature of the cooling water will be supplied at 298 K going forward. Predict the effluent concentration of A for coolant flowrates of 50 to 150 L/min. Justify your assumptions. (10)
 - b. Given more time and broader scope, we would take you to determine the optimal coolant flowrate to minimize penalties on the concentration of A in the effluent, while ensuring that the CSTR's temperature remains below 368 K.

Provide a brief overview of your approach to tackling this task. Discuss any limitations that would prohibit prescriptive analytics. (5)

REPORT FORMAT

Your report should conform to the following guidelines:

- A4 format, 11-point Calibri font 1.1 x line spacing.
- Page Limits will be strictly enforced. Any work submitted more than the limit will not be marked.
- It must be a pdf document.
- Last Page to be a standard UCT plagiarism declaration, signed by you.

NB. If you make use of any AI tools, you should make a declaration to that effect and explain exactly where and how AI was used. Failure to do so may result in being penalised if discovered.

SUBMISSION

Your report (online PDF) and python file should be uploaded to Amathuba by

17h00 on Friday 17th October 2025

Lateness will be penalised as follows:

• 0 – 1 hour: 5% penalty • 1 – 2 hours: 10% penalty	• 2 – 24 hours: 20% penalty • Beyond 24 hours: 100% penalty
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